

Posterior Amplification in CS-128: Equivalent-Key Recovery and a Fresh-Message Forgery

2.3 Million Reference Signatures on One Fixed Key

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Provenance and review status. This work was implemented and tested during an AI-run audit by OpenAI Codex using Daybreak Blue at maximum reasoning effort. The public report by Yijian Liu on behalf of Xianhui Lu established priority for the underlying CS compressed-signature leakage and its claimed key-recovery consequence [2, 3]. Their attack code and transcripts were not available. This note gives a self-contained derivation and independent implementation, adds a verifier-conditioned approximate-posterior score and a deterministic public completion procedure, and records a concrete fresh-message forgery from the recovered equivalent key. It does not claim priority for the general key-recovery consequence of the leakage. The derivation, implementation, experiment, and manuscript were produced by the AI. At the date above, no human cryptanalyst has independently verified the argument, implementation, measurements, or novelty assessment. This is a preliminary disclosure for human review and reproduction.

Abstract

CS is a Fiat–Shamir-with-aborts lattice signature whose full construction compresses the public product and omits the final response vector from the signature. Verification publicly reconstructs a noisy vector

$$z'_2 = y_2 - c_r(b_0 - s_2) + \nu,$$

where b_0 and s_2 belong to the signing key and c_r contains six hidden bimodal signs. The accepted $z_0 = y_0 + c_r$ is statistically correlated with those signs. Liu and Lu used this fact to report CS-128 key recovery from 2^{27} signatures.

We give an independent implementation and refinement using $2,300,000 < 2^{21.14}$ signatures, about 58 times fewer, on one fixed CS-128 key. One million signatures from the unmodified reference signer give a rough public estimate of $d = b_0 - s_2$. For every later transcript we enumerate the 2^6 possible hidden sign vectors and combine the unconditioned $D_{5,5}$ mask likelihood from z_0 with the residual energy $\|z'_2 + c_r \hat{d}\|^2$ in an approximate posterior score. Regression of public z_1 on the resulting approximate-posterior mean over 1.3 million further signatures gives a ternary candidate differing from the originally sampled s_1 at two coefficients. A deterministic public completion ladder changes one coefficient and obtains a compatible equivalent signing key.

The completed candidate u determines compatible values s_2^* and b_0^* . We serialized the resulting equivalent signing key, signed a message absent from the training set, and obtained acceptance from the reference CS-128 verifier. The experiment ran on a four-core Intel Core i5-6500; Section 7 reports its measured time. The attack follows the normative signing and verification equations and demonstrates an end-to-end attack through the specified design on the tested key. Success probability over independently generated keys has not been measured.

1 Introduction

CS (“Compact-Simple”), a signature candidate in the Next-generation Commercial Cryptographic Algorithms Program (NGCC), is a module-lattice construction with bimodal rejection sampling and two compression mechanisms [1]. The public key contains only the high part b_1 of the module-LWE product

$$b = A's_1 + s_2 = 2^\beta b_1 + b_0,$$

while a signature transmits z_0 , z_1 , a hint, and a Fiat–Shamir seed. The verifier reconstructs the missing z_2 component from those values. This saves public-key and signature bandwidth, but the reconstruction retains a product between secret low information and the hidden signs chosen by the iterative bimodal sampler.

Liu and Lu publicly identified this leakage and reported complete CS-128 recovery from 2^{27} signatures generated in 3.25 hours on 16 cores [2]. The public NGCC tracker verified the algebraic identity but classified the statistical recovery as a lead because no code or transcript was supplied [3]. Their report estimates the hidden bimodal aggregate from (z_0, c) , regresses the reconstruction to obtain low secret information, and then uses a cross moment to obtain s_1 .

This note independently implements and strengthens that attack. Its main observation is that the public reconstruction z'_2 should be used twice: first as the regression target for a rough low-part estimate and then as side information when inferring each transcript’s hidden signs. CS-128 has only

$$N = \left\lceil \frac{\tau}{\tau'} \right\rceil = \left\lceil \frac{23}{4} \right\rceil = 6$$

hidden block signs. Exact enumeration therefore costs only 64 likelihood evaluations per signature. Even a rough estimate of the low secret term gives a strong 768-coordinate energy test for those 64 choices. The resulting approximate-posterior score is informative enough that regression on z_1 , followed by the deterministic public completion ladder, constructs an equivalent signing key in the experiment below.

The candidate-producing path and its stopping decision do not read the target secret. A candidate s_1 is accepted only if every coefficient of the public rounded product can be completed by a ternary s_2 , after which the equivalent serialized key must produce a reference-verifier-accepted signature on a fresh message.

The attack has the following concrete properties.

1. Every estimator input is reconstructed from the public key and ordinary signatures.
2. The sample complexity is $2,300,000 < 2^{21.14}$ signatures, about 58.4 times below the previously reported 2^{27} experiment.
3. Statistics are accumulated in streaming form; the largest per-worker state file is 110,608 bytes, and no signature corpus is retained.
4. The final completion check and the equivalent-key construction use only the public key.
5. The final result is an accepted fresh-message forgery, rather than a coefficient-match diagnostic alone.

Section 2 isolates the relevant CS equations. Section 3 proves the public leakage identity. Sections 4 and 5 give the two-stage estimator. Section 6 describes public correction and equivalent-key construction. Section 7 reports the full-parameter experiment and conformance checks. Sections 8 and 9 discuss scope and repair.

2 CS-128 notation and signing equations

Let

$$\mathcal{R}_q = \mathbb{Z}_q[x]/(x^n + 1), \quad (n, q, k, \ell) = (256, 32257, 3, 3).$$

Products below are negacyclic ring products, componentwise across module coordinates where appropriate. We use centered representatives whenever a small integer value is required. The remaining relevant CS-128 parameters are

$$(\tau, \tau', N) = (23, 4, 6), \quad 2^\beta = 32, \quad \alpha = 2016. \quad (1)$$

The secret vectors $s_1 \in \{-1, 0, 1\}^{\ell \times n}$ and $s_2 \in \{-1, 0, 1\}^{k \times n}$ satisfy

$$b = A's_1 + s_2 = 2^\beta b_1 + b_0 \pmod{q}, \quad (2)$$

where the public key is (ρ_0, b_1) , $A' = \text{ExpandA}(\rho_0)$, and b_0 is stored only in the secret key. Define the effective hidden low term

$$d = b_0 - s_2 = A's_1 - 2^\beta b_1 \pmod{q}. \quad (3)$$

Although Equation (3) becomes public once s_1 is known, d is secret during the statistical phase.

For one signing attempt, the signer samples independent mask vectors

$$y_0 \leftarrow D_{5,5}, \quad y_1 \leftarrow D_{10,9}^\ell, \quad y_2 \leftarrow D_{6,9}^k.$$

Here a coefficient of $D_{a,b}$ is the difference of two sums of a independent uniform b -bit integers. The Fiat–Shamir challenge c has $\tau = 23$ signed nonzero coefficients. The sampler processes the challenge support in six consecutive blocks $S_0, \dots, S_5$ of sizes 4, 4, 4, 4, 4, 3. It chooses a hidden sign $\varepsilon_g \in \{+1, -1\}$ for each block, subject to the specified rejection test, and forms

$$r = c_r = \sum_{g=0}^5 \varepsilon_g c|_{S_g}. \quad (4)$$

Thus r has the public support and coefficient magnitudes of c , while six signs are hidden. An accepted response satisfies

$$z_0 = y_0 + r, \quad z_1 = y_1 + r s_1, \quad z_2 = y_2 + r s_2. \quad (5)$$

Only z_0 and z_1 are encoded. The hint lets the verifier reproduce the high part of the commitment w and reconstruct a bounded vector denoted z'_2 below. All quantities needed to compute z'_2 are public in a verification transcript.

3 The verifier-visible leakage identity

We first restate the correctness calculation in a form useful for estimation. Let $\text{LowBits}_h(w, \alpha)$ denote the centered low part associated with the hint decomposition, and let $\text{LSB}(w)$ denote the parity lift used by CS. Define

$$\nu = \frac{-\text{LowBits}_h(w, \alpha) + \text{LSB}(w)}{2}. \quad (6)$$

The parity conventions make the numerator even coefficientwise.

Lemma 1 (Public reconstruction). *For every honestly generated accepted signature, Algorithm 12 publicly reconstructs*

$$z'_2 = z_2 - rb_0 + \nu = y_2 - rd + \nu \pmod{\pm q}. \quad (7)$$

Proof. The signing hint is the difference between the high parts of w and $w - 2z_2 + 2rb_0$. In verification, substitution of $z_0 = y_0 + r$ and $z_1 = y_1 + rs_1$ into the public linear expression gives

$$v = w - 2z_2 + 2rb_0 \pmod{2q}.$$

The term involving the original challenge has the same parity as r , because changing a challenge sign changes a coefficient by two; it therefore vanishes after multiplication by q modulo $2q$. The hint recovers the high part of w , and the parity check recovers its least significant bit. Algorithm 12 consequently computes

$$\frac{\alpha \text{HighBits}_h(w) + \text{LSB}(w) - v}{2} = z_2 - rb_0 + \frac{-\text{LowBits}_h(w, \alpha) + \text{LSB}(w)}{2}.$$

Using $z_2 = y_2 + rs_2$ and $d = b_0 - s_2$ proves Equation (7). $\square$

Equation (7) is the design-level leakage channel. The verifier needs z'_2 for its norm check, so the attacker obtains a noisy linear observation of rd . The sign vector r is hidden, but it is statistically visible through $z_0 = y_0 + r$. The first stage obtains a rough d using only that weak visibility. The second stage feeds the rough d back into Equation (7) to infer r much more accurately.

The scale used by the second-stage likelihood is also explained by public parameters. A $D_{6,9}$ coefficient has variance

$$\text{Var}(D_{6,9}) = \frac{6(2^{18} - 1)}{6} = 262143. \quad (8)$$

Treating the centered hint low part as approximately uniform contributes about $\alpha^2/48 = 84672$ after division by two. Their sum is about 346815. We therefore fix the public likelihood parameter $\sigma^2 = 350000$ from public parameters; no fitted secret scale is needed.

4 Stage one: a rough low-term estimate

Let $p_0(a) = \Pr[Y_0 = a]$ be the exact one-coefficient probability mass function of $D_{5,5}$. It is computed once by integer convolution of five uniform 5-bit variables with five negated copies. For a transcript t and one challenge block S_g , define

$$\lambda_{t,g}(+1) = \sum_{i \in S_g} \log p_0(z_{0,t,i} - c_{t,i}), \quad (9)$$

$$\lambda_{t,g}(-1) = \sum_{i \in S_g} \log p_0(z_{0,t,i} + c_{t,i}). \quad (10)$$

Ignoring the final joint rejection event gives the public soft sign

$$m_{t,g} = \tanh\left(\frac{\lambda_{t,g}(+1) - \lambda_{t,g}(-1)}{2}\right), \quad (11)$$

and a ring element $\hat{r}_t$ whose coefficient at $i \in S_g$ is $m_{t,g}c_{t,i}$ and whose other coefficients are zero.

We diagonalize negacyclic multiplication with the twisted Fourier transform. For each frequency f and public reconstruction component a , centered least squares gives

$$\hat{d}_{a,f} = -\frac{\sum_t \overline{(\hat{r}_{t,f} - \bar{r}_f)}(z'_{2,t,a,f} - \bar{z}'_{2,a,f})}{\sum_t |\hat{r}_{t,f} - \bar{r}_f|^2}. \quad (12)$$

An inverse twisted transform returns coefficient estimates, which we round to the small centered range allowed by Equation (2). This estimate is deliberately crude. In our one-million-signature reference pass it had mean absolute error 1.665, root mean square error 2.080, and maximum error 6.170 against the diagnostic value of d ; only 155 of 768 coefficients rounded exactly. Its purpose is to score hidden sign vectors, where the energy is accumulated over all 768 reconstruction coordinates.

The refinement score depends on this completed calibration estimate. Our streaming implementation therefore discards the calibration transcripts and applies the score to a disjoint second pass. Reusing the first million signatures would instead require retaining those transcripts until calibration finishes.

5 Stage two: verifier-conditioned approximate-posterior amplification

For each new signature, enumerate $\varepsilon = (\varepsilon_0, \dots, \varepsilon_5) \in \{\pm 1\}^6$ and construct r_ε as in Equation (4). Assign the log weight

$$\log W_t(\varepsilon) = \sum_{g=0}^5 \lambda_{t,g}(\varepsilon_g) - \frac{\|z'_{2,t} + r_\varepsilon \hat{d}\|_2^2}{2\sigma^2}, \quad \sigma^2 = 350000. \quad (13)$$

The first term is the exact unconditioned $D_{5,5}$ mask likelihood used in stage one. For the correct sign vector and an accurate $\hat{d}$, Equation (7) makes the second term the energy of $y_2 + \nu$; a wrong sign vector adds a structured convolution by d . Because this likelihood intentionally omits rejection-induced dependence, W_t is an approximate posterior score rather than an exact posterior probability. Its normalized mean

$$\tilde{r}_t = \frac{\sum_\varepsilon W_t(\varepsilon) r_\varepsilon}{\sum_\varepsilon W_t(\varepsilon)} \quad (14)$$

is computed by a log-sum-exp normalization. Its inputs are the public transcript, public parameters, and the stage-one estimate.

Now use the other public response in Equation (5). At each frequency f and secret-vector component j , centered least squares gives

$$\hat{s}_{1,j,f} = \frac{\sum_t \overline{(\tilde{r}_{t,f} - \tilde{r}_f)}(z_{1,t,j,f} - \bar{z}_{1,j,f})}{\sum_t |\tilde{r}_{t,f} - \tilde{r}_f|^2}. \quad (15)$$

After the inverse twisted transform, every coefficient is rounded and clipped to $\{-1, 0, 1\}$.

The likelihood in Equation (13) is intentionally simple: it treats the residual as spherical and does not model rejection-induced dependence. The experiment therefore evaluates the approximate-posterior score through the resulting public-key recovery and forgery, without treating a hidden-sign diagnostic as part of the attack evidence.

The status of the argument is consequently mixed but explicit. The public-reconstruction identity in Equation (7), the completion predicate, and the equivalent-key lemma are algebraic

consequences of the specified algorithms. The spherical likelihood and $\sigma^2 = 350000$ are modelling choices. The public ladder’s first successful sample count is an empirical result on one fixed key, rather than a concentration bound or a theorem over the key-generation distribution.

6 Public correction and equivalent-key construction

Direct rounding need not be perfect. The public key supplies an exact and inexpensive completion predicate. For a ternary candidate $u \in \{-1, 0, 1\}^{\ell \times n}$, define

$$I(u) = \# \{ (a, i) : \nexists e \in \{-1, 0, 1\} \text{ such that } \text{HighBits}_{2^\beta}((A'u)_{a,i} + e) = (b_1)_{a,i} \}. \quad (16)$$

This uses the exact submitted rounding rule, including boundary representatives. Every valid signing key has $I(s_1) = 0$. Whenever $I(u) = 0$, one admissible e at every coordinate forms a ternary vector s_2^* and

$$b_0^* = A'u + s_2^* - 2^\beta b_1 \quad (17)$$

forms its matching low part. Thus a zero of I is already an equivalent signing key; it does not have to equal the originally sampled key.

The implementation contains a deterministic public fallback ladder for runs in which direct rounding does not give $I(u) = 0$. It first tests scales from 0.7 through 1.3 in increments of 0.0002, rounding and clipping after each scale. It then ranks coefficients by the increase in squared error to the closest alternative ternary value and tests: both alternatives at each of the first 64 positions; all two-position alternatives among those 64 positions; and all three-position alternatives among the first 24 positions. The coefficient stage contains at most

$$2 \cdot 64 + 4 \binom{64}{2} + 8 \binom{24}{3} = 24,384$$

public candidates. Every decision uses only the continuous public estimate and $I(u)$. This is also the candidate ladder used by the 100,000-signature stopping rule in Section 7; the stopping decision is the public condition $I(u) = 0$ followed by a verifier-accepted fresh-message signature. Secret-coefficient diagnostics do not choose the stopping point or a candidate.

Before serialization, b_0^* is formed from the standard representative of $A'u + s_2^* \bmod q$ using the submitted implementation’s exact centered/`LowBits` representative convention. The code applies `Power2RoundVector`, then checks its high part coefficientwise against b_1 . To serialize an equivalent secret key, choose an arbitrary fresh secret seed K , copy the public matrix seed ρ_0 , compute $tr = H(pk)$, and encode $(u, s_2^*, K, tr, \rho_0, b_0^*)$. The submitted byte format additionally stores the public high part b_1 (named `t1` in the code), which is copied from the public key. The reference signer can then sign arbitrary messages for the original public key.

Lemma 2 (Equivalent-key completion). *If a ternary candidate u satisfies $I(u) = 0$, the public completion procedure constructs a well-formed CS-128 signing key for the target public key.*

Proof. Equation (16) supplies s_2^* coefficientwise and Equation (17) supplies the corresponding low part. These values satisfy the same public rounding relation as the target key. The candidate u and the completed s_2^* have the required ternary form. The public seed ρ_0 , public high part b_1 , and $H(pk)$ are copied, while the signing-randomness seed K is freely chosen. Encoding these values with b_0^* therefore gives a signing key satisfying the CS-128 key relation. Whenever the signing algorithm returns a response, the specified correctness calculation consequently verifies it under the target public key. $\square$

In the experiment below, the deterministic public candidate ladder produced a vector with $I(u) = 0$. The resulting key signed a fixed message absent from every signing-oracle query, and Algorithm 12 accepted the signature under the target public key.

The completion lemma concerns the specified algebraic key relation. Its byte-level realization uses the submitted `Power2RoundVector` and `skEncode` routines, then checks the result with `CS_Verify`. The experiment checks both layers.

7 Full-parameter experiment

7.1 Setup

The experiment used the submitted CS-128 parameter set and reference source. One fixed, deterministically generated test key was shared by all worker streams. Messages were distinct 16-byte counter-and-worker strings. Each worker accumulated Fourier moments independently; the small state files were merged without retaining signatures. Ground-truth values were recorded only to report estimator accuracy. They were not inputs to the approximate posterior scores, regressions, candidate ordering, public completion predicate, equivalent-key construction, or verifier decision.

For exact reproduction, key-generation seed byte i and worker- w seed byte i were respectively

$$0x51 + 7i \pmod{256}, \quad 0xa7 + 11i + 37w \pmod{256}.$$

The x86-64 host encoded integers in little-endian order. The first eight message bytes held query index j ; the final eight held

$$j \text{ xor } 0x43532d313238 \text{ xor } (w \ll 48).$$

The worker sets $w \in \{1, 2, 3, 4, 101, 102, 103\}$ are disjoint, and multiplication by 37 permutes bytes modulo 256, so all seven DRNG seeds differ.

The forgery target was the 42-byte ASCII message **CS-128 recovered-key fresh-message forgery**. Its length alone distinguishes it from every 16-byte signing query, so freshness does not depend on a probabilistic collision argument.

The calibration pass generated 250,000 signatures per worker through the submitted reference `CS.Sign` path, using worker streams 1–4. Refinement used the disjoint streams 101, 102, and 103 in that fixed order. Streams 101 and 102 contributed 500,000 signatures each; stream 103 stopped after 300,000 because the attack succeeded. A cumulative state was frozen every 100,000 signatures. At each point the deterministic public candidate ladder was run; the attack transcript ends at its first verifier-accepted fresh-message signature. Section 7 reports that checkpoint and its immediately preceding failure. Secret diagnostics do not select the endpoint. Every signature in those states was produced by the same unmodified reference signer and sampler, and no message or randomness stream was reused between the two passes. The collector linked copies of the submitted `cs.c` and `sampling.c` files directly. Those copies differ from the official archive only in CRLF line endings; their line-ending-normalized SHA-256 hashes match the official files and are recorded below. No alternative sampler was used, and verifier acceptance is not used as evidence of distributional identity. The forged signature was likewise produced by the submitted reference signer and checked by its reference verifier.

For transparency, the collector has no sampler instrumentation, hidden-sign trace, or fast mode. Its refinement stage refuses to run without the public stage-one estimate. A separate deterministic algebra test compares the optimized Gram-matrix evaluation of all 64 residual energies with their direct 768-coordinate evaluation; all tested scores agree exactly. In a negative-control build,

every ground-truth array supplied to the reporting routine is replaced by unrelated fixed values immediately after the public states are merged. The first successful checkpoint, correction index, completed candidate, and accepted fresh-message forgery are unchanged. This control confirms that the diagnostic comparisons do not drive recovery or completion.

7.2 Results

Table 1 summarizes the attack on a four-core Intel Core i5-6500 desktop with 16 GB of memory, Linux 7.0.0, GCC 13.3.0, and `-O3 -fno -fcommon -std=gnu11`. The managed execution environment scheduled the workers below the machine’s full four-core capacity, so we report both aggregate user CPU and observed elapsed time. The final worker’s elapsed time includes deliberate pauses at the 100,000-signature gates while the public tests ran. The public-ladder row is a complete replay of all 13 tested refinement prefixes, including equivalent-key encoding, fresh signing, and reference verification at the successful prefix.

Table 1: CS-128 full-parameter attack results.

Phase	Signatures	User CPU	Elapsed	Principal result
Reference calibration	1,000,000	28.18 min	35.15 min	$\hat{d}$ MAE 1.665, RMSE 2.080
Approx. posterior refinement	1,300,000	37.50 min	74.53 min	public transcript processing
Public ladder replay	0	0.12 min	0.25 min	depth one; $I(u) = 0$; verifier accepts

The final diagnostic record was

```
approx_posterior_s1_errors=2/768
approx_posterior_s1_maxerr=0.525291
public_correction_depth=1 index=287
public_key_consistency_invalid=0/768
equivalent_key_fresh_forgery=ACCEPT
```

Direct ternary rounding differed from the originally sampled s_1 at two of 768 coefficients. This comparison is diagnostic only. The successful path used the raw regression values, ternary rounding, and the fixed public predicate in Equation (16). The scale sub-ladder found no valid key; the coefficient sub-ladder changed index 287 and obtained $I(u) = 0$ at depth one. It did not fit or apply a secret-derived scale, and the resulting equivalent s_1 need not equal the originally sampled one.

The stage-one workers each retained 79,888 bytes and the refinement workers each retained 110,608 bytes. Consequently, the attack can consume signatures online and discard them immediately. Its total sample count is 2,300,000, or about $2^{21.13}$, compared with 2^{27} in the earlier report. Signature collection consumed 3,940.53 seconds of aggregate user CPU and 6,580.46 seconds (109.67 minutes) of observed staged elapsed time, including the checkpoint pauses.

The frozen public ladder tested refinement prefixes in increments of 100,000. It rejected every prefix from 100,000 through 1,200,000 and first accepted at 1,300,000. Thus 2.3 million total signatures are the first success at this resolution for the fixed key, calibration budget, estimator, and candidate ladder. This is not a general minimality result: the one-million calibration budget was not varied, intermediate counts were not tested, and another public completion algorithm might succeed from an earlier prefix.

7.3 Reproducibility record

The principal artifact hashes are listed in Table 2. The evidence manifest fixes every streaming checkpoint and timing record; the ladder and result files record the public stopping decision, completion, and accepted forge.

Table 2: SHA-256 hashes of the attack source and final evidence.

Artifact	SHA-256
<code>leak_reference.c</code>	dcafca8a30fe43f656dcd71b701b9c61 78a8dffaf1d75501e6e2a22926a4ea3c
Official CS.zip archive	c790d31cd4a288990f3d692381ed721a 06641938a02dfc2e0435b7d323475cef
Submitted <code>cs.c</code> and <code>sampling.c</code> (LF-normalized)	ac771cc2f138158680d1504f92bf2398 c3f8fc85b9a6137c8cebcaa18bf96ff0 bfd6abad31099e2a045644d500d3d1df 4f90ae98d50a5c10ebf0a79b0416f541
<code>allref_coarse.raw</code>	45e24ce4106e3c32772674dd7ad3d7ff 58701fe3dfdc55a8de86cc9b5b15fb15
evidence/SHA256SUMS	91ce157728b2c5b794c7d3e06768d624 d9273566d302e3de2dab6ec3318ebce4
<code>public_ladder.txt</code>	ea86e153f5a631529470142317fa212b 5236bcbbc78a2d05489f741682a10131
<code>cs128_public_ladder_</code> <code>result.txt</code>	f8527834c8fe0d4aac07a687f6657a37 cf6c03928b4c09181a95af56ceeba28d

The frozen states reproduce rejection at all twelve earlier 100,000-signature refinement checkpoints, the depth-one public correction at 1.3 million refinement signatures, zero public-key violations, and the accepted forgery when merged by the listed source.

8 Security impact and scope

For the tested key, the result is a complete specification-conforming many-query attack: it recovers an equivalent signing key and produces a valid signature on a fresh message. It uses the normative public-key compression, iterative bimodal sampler, hint equation, and verifier reconstruction in Algorithms 10–12. The leakage identity is therefore a design property; repairing rANS canonicality, signature-length handling, memory safety, or another implementation defect does not remove Equation (7).

The final key is equivalent rather than necessarily byte-identical to the original key. This distinction has no protective value: it signs arbitrary messages under the victim public key. The transcript messages can be chosen by the attacker, and the final message is fresh. The experiment therefore satisfies the forgery condition for this fixed public key. Because only one generated key was tested, it does not estimate the probability of success in the full EUF-CMA experiment over key generation.

This experiment evaluates CS-128 only. CS-256 and CS-512 have different ring dimensions, block counts, mask parameters, and compression parameters. The same algebraic identity and approximate-posterior construction apply, but this note makes no laptop sample-complexity claim for those sets. The result also does not depend on the separate challenge-sign-blind search attack recorded for CS [3].

The attack is statistical and its success probability across independently generated keys has not been measured. The experiment uses one fixed key and deterministic streams, so it does not measure key-averaged reliability. It is nevertheless a complete submitted-parameter experiment on that key: it uses public transcript processing, public-key-only completion, and an accepted fresh-message forgery.

9 Repair considerations

The failure comes from combining three facts: r is only weakly hidden by a narrow mask in z_0 , the public-key low part is secret, and verification exposes a noisy linear function of r times that secret low information. Increasing a norm bound or changing the byte encoding does not alter this channel.

A repair needs a new construction and security analysis. At minimum, the verification transcript must cease to reveal a usable relation of the form

$$\text{public reconstruction} = \text{independent noise} - r \cdot \text{secret low term}.$$

Possible design directions include publishing enough of the rounded relation that the omitted term is public, transmitting and checking a response component whose masking distribution is proved independent of the key, or redesigning the hint and compression together. Each choice changes sizes, correctness margins, or underlying assumptions. Reparameterizing the current estimator alone is not a repair: the attack exploits the scheme’s actual joint transcript distribution.

10 Conclusion

CS-128’s compressed verifier exposes a noisy product between six hidden bimodal signs and secret low key information. A one-million-signature public regression gives a rough model of that term. Enumerating only 64 possibilities per later transcript then turns the verifier’s own reconstructed vector into a strong approximate-posterior score for the hidden signs. After 1.3 million further signatures, the deterministic public ladder changes one rounded coefficient, constructs an equivalent key, and produces a reference-verifier-accepted fresh-message signature. The complete experiment uses 2.3 million signatures. On the four-core i5-6500, collection consumed 65.68 minutes of aggregate user CPU and 109.67 minutes of observed staged elapsed time, including deliberate checkpoint pauses in the managed execution environment.

References

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